# MCO-1D

## Operator Manual

How to fly, talk to, and work with your ornithopter drone.

| | |
|---|---|
| **Document** | MCO-1D_Operator_Manual_v21 |
| **Revision** | 21 — August 2026 |
| **Changes** | Wing coordination, forest flight, obstacle sensing, ceiling exploration |
| **Audience** | Pilot / Operator (no engineering background required) |
| **Companion** | MCO-1D Commissioning Manual v20 |

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# 1 What This Manual Is For

This manual tells you — the pilot — everything you need to operate the MCO-1D ornithopter drone. It assumes the aircraft has been built and commissioned. No engineering knowledge required.

## What this manual covers:

- Starting up the ground station on your laptop
- Talking to the aircraft in plain English
- Arming, hovering, cruising, and landing
- The complete throttle lever guide
- Braking, deceleration, and J-hook
- How the aircraft learns and improves continuously
- Wing coordination and clap-and-fling physics
- Forest flight, gap negotiation, and obstacle avoidance
- Autonomous learning sortie programme
- Altitude ceiling exploration (airspace-gated)
- After-flight shutdown and learning preservation
- Troubleshooting

> ◆ **NOTE** This manual contains no Python, no command names, no jargon. Everything is in plain English. For engineering detail see the Commissioning Manual.

## Aircraft at a Glance

| | |
|---|---|
| **Aircraft** | MCO-1D metachronal cascade ornithopter SAR drone |
| **Wings** | 8 — four tandem pairs W1(front) through W4(rear), independently controlled |
| **Weight** | ~4.1–5.1 kg assembled and flying |
| **Stall speed** | 4.85 m/s (~9.4 knots) |
| **Maximum speed** | 28 m/s (~54 knots) |
| **Flapping** | 0.3–3.0 Hz hard limit, 89.5–115° stroke arc depending on coordination mode |
| **Flight computer** | Raspberry Pi on the aircraft |
| **Ground station** | Your laptop — connects over Wi-Fi |
| **Pilot interface** | Plain English conversation |

# 2 Equipment You Will Need

Before your first flight, make sure you have everything below.

## Aircraft side (pre-commissioned)

- MCO-1D assembled and commissioned per the Commissioning Manual
- Charged battery packs (one minimum; three for a learning session)
- Clear open area minimum 20×20m, no overhead obstructions
- Landing pad 1×1m, flat and clearly marked

## Ground station side (your laptop)

- **Laptop or tablet** — any modern browser, Windows / macOS / Linux
- **Python 3.9+** installed
- **Two packages:** `pip install aiohttp anthropic`
- **Anthropic API key** from console.anthropic.com — set as environment variable
- **Three files:** mco_ws_bridge.py · mco_api_proxy.py · mco_ground_station_launcher.sh
- **Interface:** mco_ai_ground_station.jsx in Claude Artifacts or local server

> ■ **CAUTION** Your API key is personal. Never share it or put it in a file. Keep it in an environment variable only.

# 3 Starting the Ground Station

Three components must all be running before you open the interface. The launcher handles the first two automatically.

## Starting up — step by step

**1** **Power on the aircraft**

Connect the battery. Wait for the solid green status light (~10-15 seconds).

Do not arm until the light is solid green.

**2** **Connect laptop to the aircraft network**

Join the Wi-Fi. Confirm: open a terminal and type `ping 192.168.7.1`. You should see replies.

Replace 192.168.7.1 with your aircraft's IP if different.

**3** **Set your API key**

In a terminal:

None

```
export ANTHROPIC_API_KEY=sk-ant-...
```

**4** **Run the launcher**

In the same terminal, navigate to the MCO-1D files folder and run:

```
bash mco_ground_station_launcher.sh 192.168.7.1
```

**5** **Open the interface**

Go to claude.ai, attach mco_ai_ground_station.jsx, ask Claude to render it.

**6** **Check the connection badge**

Top-right: green DISARMED = ready. Red OFFLINE = check launcher and aircraft.

■ **CAUTION** If OFFLINE after 30 seconds: check IP address, aircraft light, restart launcher.

# 4 The Conversation Interface

The ground station is a conversation. You type plain English. The AI understands your intent, operates the aircraft, and tells you what happened.

## What you see

| | |
|---|---|
| **Six steam gauges (top)** | Airspeed, attitude, altitude, vertical speed, power, battery. Analogue instruments with coloured arc |
| **Status strip (below gauges)** | Armed/safe indicator, fault count, flight state, discovery on/off, ALSS progress, ceiling probe altitud |
| **Conversation panel** | Your messages right, AI replies left. Red alerts when action needed. |
| **Input box** | Type and press Enter. Shift+Enter for new line. |

## How to talk to the AI

Speak naturally — the AI understands intent, not just keywords.

*"Ready to fly"*
→ AI runs all pre-flight checks.

*"Arm it"*
→ AI confirms once, then arms. Badge turns amber.

*"It's drifting left"*
→ AI checks attitude hold, diagnoses, adjusts.

*"Enable clap and fling"*
→ AI opens the clap gate and explains what will happen.

*"Going into the forest"*
→ AI enables obstacle avoidance and explains the sensor ring.

*"The gap is 40 centimetres"*
→ AI selects passage strategy and explains it.

*"Land here"*
→ AI descends to the surface vertically with position hold.

*"We're done"*
→ AI saves learning and disarms.

◆ **NOTE** The AI watches the instruments continuously. It speaks up about faults, low battery, wing guard interventions, and LEV risk at altitude.

# 5 Pre-Flight Checks

Before every flight, run a pre-flight check. Just tell the AI you are ready and it does everything.

> *"Ready to fly"*
> → AI checks: faults, battery (must be above 21.5V), attitude hold, hover mode, airspeed sensor, GPS lock, rope health, wing guard state.

## Common failure causes

**Battery below 21.5V** — Charge the battery before flying.

**GPS not locked** — Wait 60-90 seconds in the open, away from buildings.

**Fault from previous flight** — Ask the AI what the fault was.

**Airspeed sensor blocked** — Check nothing covers the pitot tube at the nose.

## Arming

Once checks pass, say "arm it". AI confirms once, then arms. Badge turns amber. Wings can move from this point.

■ **CAUTION** Stand clear of all eight wings once armed. 2-metre clearance minimum.

# 6 Normal Flight Operations

Two flight modes: hover (below ~3 m/s) and fixed-wing cruise (above ~5 m/s). Transition is automatic.

| | |
|---|---|
| HOVER (below 3 m/s) | Lever = altitude. Joystick = horizontal velocity. Release stick = hold GPS position. |
| FIXED-WING (above 5 m/s) | Lever = thrust/braking. Joystick = bank, pitch, yaw. Smooth blend between modes at 3-5 m/s. |

## Taking off

"Take off" or "let's go" — AI increases throttle until clear of ground, holds hover height.

## Turn tightening

At hard roll inputs in cruise, the aircraft automatically uses VIFF — differential thrust vectoring — to produce tighter turns. Invisible to the pilot. The aircraft simply responds more effectively to aggressive stick inputs.

## Landing

"Land" or "bring it down" for normal landing. "Steep approach" for confined spaces. For complex surfaces (slopes, rubble, ledges) see Section 12.

◆ **NOTE** Say "we're done" after touchdown to save learning before approaching the aircraft.

# 7 The Lever — Complete Guide

The throttle lever controls the aircraft's energy state. Forward = more power. Back = less power and braking. 50% = altitude hold.

| ZONE | LEVER | WHAT HAPPENS |
|---|---|---|
| THRUST | 0–5% | Maximum burst / fast sprint. |
| SPRINT | 5–28% | High-speed sprint. Maximum thrust. |
| POWERED CRUISE | 28–45% | Normal cruise. |
| ALTITUDE HOLD | 45–55% | Holds altitude. 50% = exact hold. |
| GLIDE / RANGE | 55–75% | Efficient glide. Kestrel soaring here in good wind. |
| LIFT DUMP PARTIAL | 75–80% | Rear wings reduce lift — steeper approach. |
| LIFT DUMP FULL | 80–87% | Rear wings stop — full spoiler equivalent. |
| SHUTTLE BRAKE ½ | 87–93% | Progressive aerodynamic braking. |
| J-HOOK | 93–97% | Post-stall pitch-up to hover. Needs 20m+. |
| SHUTTLE BRAKE FULL | 97–100% | Maximum drag. ~2g deceleration. |

# 8 Braking and Deceleration

All braking modes are engaged by lever position — no separate switch needed.

## Lift dump — steeper approach

Lever 75–87% stops the rear wing pairs, reducing total lift for a steeper approach. Gradual: 75% is gentle, 87% is full spoilers equivalent.

> *"Steepen the approach"*
> → AI moves lever into lift dump zone.

## Shuttle brake — aerodynamic deceleration

Lever 87–97%: all wings pitch to high drag angles. Like a shuttlecock facing the airflow. Builds to 1.5–2g at full position. Auto-exits when slow.

## J-hook — post-stall pitch-up to hover

Lever 93–97% above 20m: rapid nose-up, hold while decelerating, transition to hover. Completes in under 3 seconds. Fails safe to shuttle brake below 20m.

■ **IMPORTANT** Emergency stop cuts all motors instantly. Only if crash is certain and people are in the flight path.

# 9 How the Aircraft Learns

The MCO-1D improves continuously while flying. Every flight hour adds to its knowledge. You do not need to do anything for this to work.

## Three learning systems run simultaneously

| | |
|---|---|
| **Discovery** | Tests small wing configuration variations during stable cruise. Keeps better-scoring variants, dis |
| **MAP-Elites archive** | Stores the best-ever configuration for every airspeed × wind × intent combination encountered. E |
| **Strategy genome** | Learns which wing mechanism combinations work best for each manoeuvre — rolls, climbs, bra |

*"That worked well"*

→ AI records a positive observation — configuration gets credit.

*"The hover feels wobbly"*

→ AI sets the learning focus to hover stability.

## Saving between flights

Learning saves automatically when you say "we're done". Everything carries forward to the next flight. Each battery cycle adds to what the aircraft knows.

■ **CAUTION** Say "save learning" before disconnecting the battery if you need to stop quickly.

# 10 Mission Focus

Tell the aircraft what to get better at. The learning system concentrates on what you describe.

| WHEN YOU SAY… | WHAT IT DOES |
|---|---|
| *"Gusty / rough air"* | Optimises for turbulence stability. |
| *"Hold position better"* | Concentrates on GPS position hold accuracy. |
| *"Smooth / camera payload"* | Minimises vibration. |
| *"Crosswind landing"* | Optimises crosswind approach. |
| *"Stay up longer"* | Minimises power — endurance focus. |
| *"Fast / top speed"* | Optimises for maximum cruise speed. |
| *"SAR / rescue"* | SAR hover and confined-space operations. |
| *"Stormy conditions"* | Storm resilience — maximum dihedral, load shedding. |
| *"Tight aerobatic turns"* | Roll authority and turn tightening. VIFF optimised. |
| *"Good updraft here"* | Kestrel soaring mode if conditions are right. |
| *"Enable clap and fling"* | Clap-and-fling hover optimisation. See Section 11. |
| *"Anti-phase cruise"* | Maximum amplitude endurance cruise. See Section 11. |
| *"Back to normal"* | Clears focus — general optimisation resumes. |

# 11 Wing Coordination and Clap-and-Fling

The MCO-1D has eight wings in four port-starboard pairs. The port and starboard wings of each pair stroke toward and away from each other continuously in flight. The geometry means their tips can approach each other at high stroke amplitudes — and with the right timing, they can be made to clap together, producing a significant aerodynamic benefit.

## The wing guard — always on

A real-time coordinate system monitors the position of both wings in each pair every control cycle. If the tips come within 50mm of each other without a validated clap in progress, the system reduces amplitude on both wings automatically. The pilot feels nothing — the aircraft simply stays within safe limits. This cannot be turned off.

## Safe amplitude depends on phase

How widely the wings can stroke depends on how they are timed relative to each other. When both wings of a pair move in the same direction at the same time (in-phase), the maximum safe amplitude is 89.5° — a conservative limit with 5° clearance. When one wing is at full downstroke while the other is at full upstroke (anti-phase, 180° offset), they are always at opposite extremes and never approach each other — the amplitude can safely reach 115°. The system manages this automatically based on the commanded phase relationship.

| | |
|---|---|
| **In-phase (ϕ=0°)** | 89.5° max amplitude — wings converge simultaneously |
| **Quarter-phase (ϕ=90°)** | ~102° max — partial separation maintained |
| **Anti-phase (ϕ=180°)** | 115° max — wings always at opposite extremes |
| **Clap gate open** | 100.5° max — validated contact at stroke reversal |

## Clap-and-fling — what it is and what it does

When flying insects need maximum lift — a heavy load, thin air, aggressive hover — many use a mechanism called clap-and-fling. The wings meet at the top of the upstroke, clapping together. As the downstroke begins, they fling apart. The fling creates a leading-edge vortex — a spiralling air mass — that the wing then sweeps through on the downstroke, capturing additional circulation. The net effect is approximately 15-25% more lift for the same power in hover.

The MCO-1D's membrane wings can flex through controlled contact at the right moment. With the clap gate open, the wing guard validates the timing of every approach — allowing contact only when both wings are converging at stroke reversal, moving at the correct speed. Unvalidated approaches are still prevented.

*"Enable clap and fling"*

→ AI opens the clap gate. Wings will touch at top of upstroke. Stroke amplitude increases to 100.5°. Hover lift improves 15-25%. Guard remains active — timing validated every tick.

*"Disable clap / normal wings"*

→ AI closes the clap gate. Normal amplitude limits restore.

*"Wing clearances / tip separations"*

→ AI reads clap_status — shows live tip separations.

*"Anti-phase cruise / maximum amplitude cruise"*

→ AI sets 180° port-starboard phase offset. 115° amplitude. ~20% more stroke volume per cycle. More efficient at cruise.

◆ **NOTE** Clap-and-fling is most useful in hover with payload, in thin air at altitude, or whenever maximum lift per watt matters. In forward flight, anti-phase wide amplitude is generally more useful.

◆ **NOTE** The wing guard is always running. You will never hear about it unless a guard event occurs — at which point the AI will mention it and explain what happened.

# 12 Forest Flight and Obstacle Avoidance

The MCO-1D has the wing geometry, the control authority, and the gap-negotiation manoeuvres to fly through complex environments — forest canopies, building interiors, dense brush, confined spaces. To do so safely, obstacle sensing must be active and the correct passage strategy selected for the gap you are entering.

## Enabling obstacle avoidance

When you say "going into the forest" or "enable obstacle avoidance", the aircraft activates its ring of time-of-flight sensors — forward, forward-left, forward-right, port, starboard, and downward — and begins computing the safest heading based on what they detect. Your stick commands remain fully effective. The avoidance system adds corrections on top of your inputs, never replacing them.

| | |
|---|---|
| **Forward sensor (>3m clear)** | Free cruise — no correction. |
| **Any sensor (1.5-3m)** | Caution — bias away from obstacle, slow down. |
| **Forward sensor (<1.5m)** | Danger — active avoidance + gap assessment. |
| **Gap detected automatically** | AI alerts: gap detected ahead. Say "go through the gap" to enter. |

## Gap negotiation — three strategies

When you say "the gap is X metres wide", the AI selects the correct passage strategy based on that width relative to the aircraft's 2.97m full wingspan.

| | |
|---|---|
| Gap > 0.55m | FLAPPING PASSAGE — wings continue flapping, upstroke held at minimum-span position as body tra |
| Gap 0.20–0.55m | SIDEWAYS PASSAGE — body rolls 90°, one side sweeps fully aft for clearance, other side maintains |
| Gap 0.12–0.20m | BALLISTIC TUCK — wings sweep fully aft and stop at neutral. Body coasts through on airspeed. Stabl |
| Gap < 0.12m | IMPASSABLE — narrower than the fuselage. |

*"The gap is about 40 centimetres"*

→ AI: gap_set_width 0.4 → ballistic tuck strategy. Explains that wings will sweep aft and stop. Asks if ready to enter.

*"Go through the gap"*

→ AI engages gap_enter. Gap negotiation begins.

*"Abort / come back"*

→ AI: gap_abort. Wings return to cruise configuration.

## Canopy approach and perch landing

The Cooper's hawk uses a flap-flap-glide pattern through forest canopy — two wingbeat bursts then a glide phase — rather than continuous flapping. This produces lower acoustic signature and better stability in turbulent canopy air. The MCO-1D uses the same pattern for canopy approach.

*"Approach through the trees / fly in under the canopy"*

→ AI engages canopy approach profile. Flap-flap-glide pattern. Decelerates to hover above the target point.

*"Land here / set it down on the ledge / land in the clearing"*

→ AI engages vertical descent with GPS position hold. Tolerates slopes up to 15°. Descends until contact or your stop command.

*"Get out / climb out / exit the space"*

→ AI engages confined exit — front wings at maximum hover power, rear wings stopped to keep span at 0.45m during the vertical climb. Expands to full span once clear.

■ **CAUTION** Obstacle avoidance requires the 6× time-of-flight sensor ring and PMW3901 optical flow sensor to be physically installed. Ask your commissioning engineer whether these are fitted before entering confined spaces.

# 13 Autonomous Learning Sorties

The Autonomous Learning Sortie System (ALSS) lets the aircraft fly, learn, and repeat with minimal human involvement. Define the flying area, start the programme, swap batteries.

## What happens

The aircraft takes off, works through a planned sequence of manoeuvres, returns, lands, and waits. Swap the battery and arm again — the next sortie adapts based on what the previous one learned.

| | |
|---|---|
| CLASS A (first 5 sorties) | Orientation. Hover at three heights, gentle cruise, rolls, climbs. Nothing aggressive. |
| CLASS B (sorties 6-20) | Exploration. Efficiency cruise, banked turns, lift dump, hard climbs, clean glide. |
| CLASS C (sortie 21+) | Performance. Full roll, VIFF turns, shuttle brake, J-hook, clap hover, anti-phase cruise. Only unlocks |

## Setting up

1. **Define the flying area**
   "Set the practice area to [lat] [lon], 100 metres radius, 50 metres high" — aircraft will not leave this cylinder.
2. **Start**
   Say "start the practice flights". AI confirms and begins.
3. **Swap batteries when prompted**
   When you hear "swap battery and re-arm", swap, arm, and the next sortie begins. Learning saves automatically between sorties.
4. **Monitor**
   Ask "how are the sorties going?" at any time.
5. **End**
   Say "stop the practice flights". AI completes current sortie, lands, saves.

■ **CAUTION** A human safety observer should be present, especially for first Class C sorties. "Abort" or "land now" to end any sortie immediately.

# 14 Ceiling Exploration

■ **IMPORTANT** Ceiling exploration may reach altitudes requiring regulatory notification or authorisation in your area. Obtain all necessary permissions before proceeding.

## What the aircraft is looking for

As altitude increases, air gets thinner. The aircraft compensates by increasing flapping amplitude and frequency. At some altitude, these reach hardware limits — that is the ceiling. The aircraft probes this boundary very slowly, holding each new altitude for 60 seconds to test stability.

## The two-step authorisation

**1** **Complete the checklist**

"Arm ceiling exploration" → AI displays the checklist and arms for 60 seconds.

- ■ Airspace authorisation obtained
- ■ NOTAM filed or authority notified
- ■ Chase observer in position
- ■ Wind below 5 m/s
- ■ Battery fully charged
- ■ Aircraft in stable cruise

**2** **Confirm to begin (within 60 seconds)**

"Confirm ceiling exploration" → probe begins. Must come within 60s of arm.

If you wait more than 60 seconds, the arm expires and you must start again.

## During the probe

Climb rate is 0.15 m/s — very slow and intentional. Allow 30-60 minutes for a complete probe. The aircraft holds each altitude for 60 seconds before attempting to climb further.

*"How high is it / ceiling status"*

→ AI gives altitude, saturation margins, LEV risk.

*"Abort ceiling test"*

→ AI descends immediately.

# 15 After Flight

Proper shutdown preserves learning and leaves hardware safe.

1. **"We're done" or "land and save"**
   AI saves all session learning and disarms. Single phrase does everything.
2. **Wait for DISARMED badge**
   Status badge returns to green. Then approach.
3. **Disconnect battery**
   Only after AI confirms disarmed and saved.
4. **Close ground station**
   Ctrl+C in the terminal stops everything cleanly.

◆ **NOTE** Disconnecting without saving: that session's learning is lost but all previous flights remain intact. Say "save learning" first if needed.

# 16 Troubleshooting

Start with the AI — it can diagnose most faults directly from the aircraft's data.

### Interface shows OFFLINE

Launcher not running or aircraft not reachable. Check: launcher running, aircraft light green, laptop on same network, ping 192.168.7.1 gets replies.

### Pre-flight fails on battery

Charge battery. Must be above 21.5V to fly.

### Pre-flight fails on GPS

Open area, away from buildings. Wait 60-90 seconds.

### Pre-flight shows a fault

"What is the fault?" — AI explains in plain English.

### Aircraft drifts in hover

"It's drifting" — AI checks attitude hold and GPS.

### Wing guard intervention reported

The guard prevented a potential wing collision. Ask "what happened" and the AI will explain what triggered it. Usually caused by aggressive roll input combined with high amplitude. Normal operation resumes automatically.

### Gap negotiation aborted

The aircraft detected it could not complete the passage safely. Ask why — AI will report sensor readings at the time of abort.

### Ceiling probe stopped unexpectedly

A flight limit was reached and the aircraft descended safely. Ask "why did it stop" — AI explains what limit was hit.

### AI doesn't understand

Rephrase simply. Say "no, I meant…" and it tries again.

### Lost connection during flight

Aircraft returns to takeoff point autonomously. Do not approach until landed and powered down.

# Appendix A Useful Phrases

Use whatever words feel natural. These are examples only.

## Starting

*"Ready to fly"*
→ Pre-flight checks.

*"Arm it / Start up"*
→ Arms after confirmation.

*"Disarm / Shut down"*
→ Disarms on ground.

## Flight control

*"Take off / Let's go"*
→ Takeoff sequence.

*"Land / Bring it down"*
→ Landing sequence.

*"Hold position / Stay here"*
→ GPS position hold.

*"Steep approach / Short landing"*
→ Lift dump for confined space.

*"Stop fast"*
→ Shuttle brake.

*"J-hook / Hard stop"*
→ J-hook if height allows.

## Wing coordination

*"Enable clap and fling"*
→ Opens clap gate. ~25% hover lift.

*"Disable clap / Normal wings"*
→ Closes clap gate.

*"Wing clearances / Tip separations"*
→ clap_status.

*"Anti-phase cruise"*
→ 115° amplitude, 180° phase. More efficient.

*"Wing guard events"*
→ wing_guard_status — recent interventions.

## Forest flight

*"Going into the forest / Confined space"*
→ Enables obstacle avoidance.

*"Obstacle status / What can you see"*
→ Live obstacle map.

*"The gap is X metres wide"*
→ Sets gap width, selects strategy.

*"Go through the gap"*
→ Begins gap passage.

*"Approach through the trees"*
→ Canopy approach profile.

*"Land here / Set it down on the ledge"*
→ Perch landing.

*"Get out / Climb out"*
→ Confined exit — minimum span vertical climb.

*"Disable obstacle avoidance"*
→ Returns to standard flight.

## Monitoring

*"How high? / Battery? / Is anything wrong?"*
→ Live telemetry.

*"How is the learning going?"*
→ Learning progress summary.

*"Why is it doing that?"*
→ AI reads telemetry and explains.

## Learning / Focus

*"Gusty today / Handle the wind"*
→ Gust tolerance focus.

*"Stay up longer"*
→ Endurance focus.

*"Turn tighter / More agile"*
→ Aerobatics focus.

*"That worked well"*
→ Records positive observation.

*"Back to normal"*
→ Clears focus.

## Autonomous sorties

*"Start practice flights"*
→ Starts ALSS.

*"How are the sorties going?"*
→ ALSS progress report.

*"Stop the practice"*
→ Ends after current sortie.

*"Abort / Land now"*
→ Immediate landing.

## Ceiling exploration

*"Arm ceiling exploration"*
→ Step 1 — checklist + 60s arm.

*"Confirm ceiling exploration"*
→ Step 2 — begins probe.

*"Abort ceiling test"*
→ Descends immediately.

## Ending

*"We're done / End of day"*
→ Saves learning and disarms.

*"Save the learning"*
→ Saves immediately without disarming.

*"Shut everything down"*
→ Saves, disarms, close launcher.

# Appendix B What the Aircraft Says

What the AI says on behalf of the aircraft, and what it means.

**"All checks pass. Ready to arm."**
Pre-flight completed normally.

**"Arming now."**
Stand clear of the wings.

**"Alert: fault detected (N faults)."**
Check fault_summary immediately.

**"Alert: battery low (X.Xv)."**
Plan to land soon. Below 20V: land immediately.

**"Clap gate open — dorsal mode."**
Clap-and-fling is active. Wings will contact at stroke reversal. Guard still validates timing every tick.

**"Wing guard intervention — W1 amplitude reduced."**
The guard prevented a potential collision on wing pair 1. Normal — amplitude auto-restored.

**"Gap ahead: ~Xm wide — say "go through the gap" to enter."**
Forward sensor detects an obstacle with clear sides. Gap passage available.

**"Alert: wing stall risk at this altitude. Descending."**
During ceiling probe: LEV collapse risk exceeded threshold. Descending safely.

**"Ceiling probe: saturation at Xm AGL."**
Hardware limit reached. Holding altitude and running discovery loop.

**"ALSS Sortie N complete. Swap battery and re-arm."**
Autonomous sortie finished. Swap battery, arm, next sortie begins.

**"Session saved. Safe to disconnect."**
All learning saved. Safe to cut battery.

# MCO-1D Operator Manual

Revision 21 — August 2026

Added: Wing Coordination and Clap-and-Fling (§11), Forest Flight and Obstacle Avoidance (§12), updated ALSS (§13) and Ceiling Exploration (§14). For hardware, assembly, and commissioning: see MCO-1D Commissioning Manual.